

Code-Layout, Readability and Reusabil

Date	1 July 2026
Team ID	XXXXXX
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the EduGenie codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Uses standard Python indentation (4 spaces) consistently throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Project uses separate app, routers, services, models, frontend, and tests folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Names such as learner_query, topic, generated_response, and quiz_score clearly communicate purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as generate_learning_response(), create_quiz(), and log_history() use meaningful names.
5	Code Comments	Inline and block comments explain logic	Yes	Key configuration, request handling, and service logic are documented with comments.
6	Modular Design	Code is split into reusable functions/modules	Yes	AI services, API routers, data logging, and frontend logic are separated into reusable modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Reusable helpers and service modules reduce repeated code across learning features.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling protect API requests and Gemini service interactions.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Gemini Learning Service	Python (Google Generative AI SDK)	Learning explanations, summaries, and feedback	High
2	Quiz Generator Service	Python (Gemini API)	Practice-question and quiz generation	High
3	Learning History Logger	Python (JSON)	Stores and retrieves learner history	High
4	Feedback Logger	Python (JSON)	Stores learner feedback from frontend	High
5	API Router Module	Python (FastAPI)	Routes learning and quiz requests	High
6	PDF Report Generator	Python (ReportLab)	Generates downloadable learning reports	Medium

Overall Code Quality Assessment:

S.No	Component / Module Name	Language / Technology
1	Gemini Learning Service	Python (Google Generative AI SDK)
2	Quiz Generator Service	Python (Gemini API)
3	Learning History Logger	Python (JSON)
4	Feedback Logger	Python (JSON)
5	API Router Module	Python (FastAPI)
6	PDF Report Generator	Python (ReportLab)